TOGGLING IN $W_0(C)$ ROOT ORDER IS CONJUGATE TO ROWMOTION: RESOLUTION OF AIM CONJECTURE 3.2 AND THOMAS'S CONJECTURE

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Abstract. Toggling is a simple operation on a partially ordered set: it adds an element to a downward-closed subset, or removes one, whenever the result is still downward closed. Toggling every element in turn, from the top down, gives rowmotion, a map studied since the 1970s. Toggles also act on numerical labels, where they become rational maps, and there it is hard to tell whether a product of toggles ever returns to where it started. Even for rowmotion, this is known only for special posets.

In 2015, the problem list of the American Institute of Mathematics workshop on Dynamical Algebraic Combinatorics conjectured that one such product has order $2h$ on the root posets of types A, B, H_3 and $I_2(m)$, where h is the Coxeter number. This product toggles the positive roots in an order determined by a Coxeter element. We resolve the conjecture. We prove that the product is conjugate to rowmotion in every finite crystallographic type and in types H_3 (for an explicit labelling of its roots) and $I_2(m)$; this also proves a 2013 conjecture of H. Thomas. In the four types of the conjecture, it follows that the product returns to the identity after $2h$ steps. Its exact order is $2h$ in types A_n ($n \geq 2$) and $I_2(m)$ with m odd, but only h in types B_n, H_3 and $I_2(m)$ with m even. The exceptional types and H_3 rely on exact computer calculations.

1. Introduction

1.1. **Toggles and rowmotion.** Let P be a finite poset. For $v \in P$, the *toggle* t_v acts on order ideals of P by adding v to an ideal I if $I \cup \{v\}$ is again an ideal, removing v if $I \setminus \{v\}$ is again an ideal, and doing nothing otherwise. Each toggle is an involution, and t_u and t_v commute unless u and v are joined by an edge of the Hasse diagram. The group they generate is the *toggle group* of P, introduced by Cameron and Fon-Der-Flaass [4] and developed systematically by Striker and Williams [26].

The most studied element of the toggle group is *rowmotion*: toggling every element once, from the top of the poset to the bottom. As a map on antichains it goes back to Brouwer and Schrijver [3], and Cameron and Fon-Der-Flaass [4] showed that it is this product of toggles. For the root poset Φ^+ of a finite Weyl group, Panyushev made three conjectures about rowmotion in its antichain form [20, Conj. 2.1]:

- it has order h if $w_0 = -1$ and order $2h$ otherwise, where h is the Coxeter number;
- its h-th power is the involution induced by $-w_0$;
- along every orbit, the average size of an antichain is $n/2$.

Armstrong, Stump and Thomas [2, Thm. 1.2] proved these conjectures by constructing a bijection, uniform in its definition, that carries rowmotion on antichains (nonnesting partitions) to the Kreweras complement on noncrossing partitions.

Einstein and Propp [10] lifted toggles to the *piecewise-linear* and *birational* settings, in which a toggle becomes a rational map on labellings of P (Section 2). Periodicity is much more delicate there. Grinberg and Roby [12, 13] proved that birational rowmotion is periodic on products of two chains and on several triangles. They also showed that it can have infinite order, already on small graded posets such as the Boolean lattice $[2]^3$ [12, §20], and they report that it has infinite

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order on the root poset of type D_4 [12, §21]. Musiker and Roby [18] gave a lattice-path formula for its iterates on products of two chains. Okada [19] proved periodicity on all minuscule posets, and Johnson and Liu [16] proved it on trapezoids, resolving a conjecture of Williams.

1.2. **Kreweras complements in the toggle group.** In his thesis [28, §4.4.1], Williams records an observation of H. Thomas. In type A_n, one of the toggle orders of Striker and Williams [26] is the order of the roots given by the c-sorting word of w_0. Thomas proposed a model of the Kreweras complement inside the toggle group of the root poset [28, Def. 4.4.4], as follows. Fix a Coxeter element c, and let $\beta_1, \ldots, \beta_N$ be the inversion sequence of the c-sorting word $w_0(c)$ of the longest element (Section 2). Toggling the elements of Φ^+ in this order defines Thomas's *nonnesting c-Kreweras complement*[1]

$$\mathrm{Krew}_c = \mathrm{Tog}_{\beta_1\beta_2\cdots\beta_N}.$$

Thomas conjectured that these maps are all conjugate to rowmotion:

Conjecture 1.1 (Thomas; [28, Conj. 4.4.5])**.** *For every finite Weyl group W, any two of the maps Krew_c are conjugate in the toggle group of $\Phi^+(W)$, and they are conjugate to rowmotion.*

In type A, Dequêne, Frieden, Iraci, Schreier-Aigner, Thomas and Williams [6] constructed an explicit equivariant bijection between noncrossing partitions under the Kreweras complement and nonnesting partitions under the toggle product $\mathrm{Tog}_{\beta_1\cdots\beta_N}$. They call this product the Kroweras complement Krow_c [6, Def. 4.2], and it is exactly the map Krew_c above. In particular Krew_c has order $2h$ in type A [6, Cor. 1.6]. Their conjugation identity [6, Prop. 4.7], combined with the Striker–Williams conjugacy of promotion and rowmotion [26], yields Conjecture 1.1 in type A. They also treated type C by folding, at the level of equivariant bijections.

1.3. **The AIM problem.** The birational lift of Krew_c is the map K_c obtained by toggling the root poset in the same order, using birational toggles. The problem list [21] of the 2015 AIM workshop on Dynamical Algebraic Combinatorics records the following.

> Birational toggles in $w_0(c)$ root orders appear to continue to have order $2h$ in type A_n. This fails, for example, in type D_4 (as does birational rowmotion).
>
> **Conjecture 3.2.** *Birational toggles in $w_0(c)$ root orders have order $2h$ in the coincidental types A, B, H_3, $I_2(m)$.*

Two features make this harder than the corresponding statement for rowmotion.

First, an inversion order of w_0 is in general not a linear extension of the root poset, so K_c is not rowmotion, and the known periodicity results do not apply to it directly. The toggle order matters a great deal. On the four-element poset $[2] \times [2]$, rowmotion has order 4, but $T_{(1,1)} \circ T_{(1,2)} \circ T_{(2,2)} \circ T_{(2,1)}$ has infinite order [12, Ex. 2.15], [10].

Second, "has order $2h$" admits two readings: that $2h$ is the exact order, or that $2h$ is a period, i.e. $K_c^{2h} = \mathrm{id}$. The two readings already differ in rank two. The second is the form in which Armstrong, Stump and Thomas [2] state Panyushev's conjecture. It is also the form of Hopkins' conjecture [15, Conj. 4.40] that birational rowmotion on a coincidental root poset has order dividing $2h$. Panyushev's own formulation [20, Conj. 2.1] specifies the exact order: h if $w_0 = -1$, and $2h$ otherwise.

[1] This is a map on order ideals of Φ^+, not the Kreweras complement $w \mapsto cw^{-1}$ on noncrossing partitions used in [2]; this is why it is renamed Krow_c in [6].

1.4. **Results.** Our first result proves Thomas's conjecture in every finite crystallographic type, for the combinatorial and the birational toggle groups alike, and extends it to the noncrystallographic coincidental types.

Theorem 1.2. *Let W be a finite Coxeter group of crystallographic type, or of type H_3 or $I_2(m)$ with Armstrong's root poset, labelled by roots as in Table 1 in type H_3. (In type $I_2(m)$ the labelling is irrelevant; see Section 7.) Let c be a Coxeter element. Then $\mathrm{Tog}_{\beta_1 \cdots \beta_N}$ is conjugate to rowmotion in the toggle group of $\Phi^+(W)$, for combinatorial, piecewise-linear and birational toggles. In particular Conjecture 1.1 holds.*

Combining Theorem 1.2 with what is known about the order of birational rowmotion settles the AIM problem.

Theorem 1.3. *Let W be of type A_n, B_n (with either the B_n or the C_n labelling of the root poset), H_3 (with the labelling of Table 1) or $I_2(m)$, and let c be any Coxeter element. Then $K_c^{2h} = \mathrm{id}$. The exact order of K_c is:*

- *$2h = 2(n+1)$ in type A_n for $n \geq 2$;*
- *$h = 2n$ in type B_n;*
- *$h = 10$ in type H_3;*
- *$\mathrm{lcm}(2, m)$ in type $I_2(m)$, that is, $2h$ for m odd and h for m even.*

Thus Conjecture 3.2 holds when "order $2h$" is read as a period. Read literally, it is false: already for $B_2 = I_2(4)$ the order is $4 = h$. Since $w_0 = -1$ exactly in types B_n, H_3 and $I_2(m)$ with m even, the orders in Theorem 1.3 follow Panyushev's dichotomy for combinatorial rowmotion.

Theorem 1.2 also explains the parenthetical remark in [21]. In type D_4 the map K_c is conjugate to birational rowmotion, so the two maps have the same order, and by [12, §21] that order is infinite.

1.5. **Method.** A toggle order induces an acyclic orientation of the Hasse diagram, and reversing all edges at a source (a *source-to-sink move*) amounts to conjugating by a single toggle. This principle goes back to the study of conjugacy of Coxeter elements [25, 11, 17], and it is stated for toggle groups in [9, §3] (see also [7]). By a theorem of Pretzel [22], two acyclic orientations are related by source-to-sink moves if and only if they have the same circulation around every cycle.

So it suffices to show that the orientation induced by $w_0(c)$ has circulation zero around every cycle, as does the orientation induced by rowmotion (Section 3). Two classical facts about c-sorting words determine the orientation of every edge:

- Dyer's theory of reflection orders [8] reduces the orientation of a cover $\gamma \lessdot \gamma + \alpha_s$ to the relative position of γ and α_s;
- a compatibility property of Reading and Speyer [24] expresses that position through the sign of a skew-symmetric form Ω_c (Section 4).

Zero circulation then follows from a parallelism property of the small squares in the root poset, together with the fact that these squares generate all cycles (Sections 5 and 6). The exceptional types E, F, G and the noncrystallographic type H_3 are finite checks (Sections 6 and 7). The orders are deduced in Section 8.

2. Preliminaries

2.1. **Root posets.** Let W be a finite Coxeter group with simple reflections $S = \{s_1, \ldots, s_n\}$, simple roots $\Delta = \{\alpha_1, \ldots, \alpha_n\}$, positive roots Φ^+, $N = |\Phi^+|$, and Coxeter number h. For crystallographic W, the *root poset* is Φ^+ ordered by $\beta \leq \beta'$ if and only if $\beta' - \beta$ is a nonnegative

integer combination of simple roots. Its covers are exactly the relations $\gamma \lessdot \gamma + \alpha_s$ with $\gamma, \gamma + \alpha_s \in \Phi^+$, and it is graded by height. For $I_2(m)$ and H_3 we use Armstrong's root posets [1], which Cuntz and Stump characterised as the unique posets with the expected enumerative properties [5]; they are described in Section 7. We write $\mathcal{H}(P)$ for the Hasse diagram of a poset P, viewed as an undirected graph.

2.2. **Coxeter elements and c-sorting words.** A *Coxeter element* is a product $c = s_{i_1} \cdots s_{i_n}$ of all simple reflections in some order; we fix such a reduced word. Consider the infinite word

$$c^\infty = s_{i_1} \cdots s_{i_n} s_{i_1} \cdots s_{i_n} \cdots .$$

Among its subwords that are reduced words for the longest element w_0, take the one whose sequence of positions is lexicographically first. This is the *c-sorting word* $w_0(c) = a_1 a_2 \cdots a_N$. Its inversion sequence

$$\beta_k = s_{a_1} s_{a_2} \cdots s_{a_{k-1}}(\alpha_{a_k}), \qquad 1 \le k \le N,$$

lists every positive root exactly once. We write $\beta \prec_c \beta'$ if β occurs before β' in this sequence.

Example 2.1. Let $W = A_3$ and $c = s_1 s_2 s_3$. Then $w_0(c) = s_1 s_2 s_3\, s_1 s_2\, s_1$, and the inversion sequence is

$$\alpha_1,\ \alpha_1 + \alpha_2,\ \alpha_1 + \alpha_2 + \alpha_3,\ \alpha_2,\ \alpha_2 + \alpha_3,\ \alpha_3.$$

This is not a linear extension of the root poset: $\alpha_1 + \alpha_2$ comes before α_2.

2.3. **Toggles.** Let $\widehat{P} = P \sqcup \{\hat{0}, \hat{1}\}$, with $\hat{0}$ below and $\hat{1}$ above every element. A *labelling* is a map $f : \widehat{P} \to \mathbb{K}$ to a field $\mathbb{K}$ (in practice a field of rational functions in the labels), with fixed boundary values $f(\hat{0}) = a$ and $f(\hat{1}) = b$. The *birational toggle* at $v \in P$ is

$$(T_v f)(v) = \frac{\sum_{u \lessdot v} f(u)}{f(v) \sum_{u \gtrdot v} 1/f(u)}, \qquad (T_v f)(x) = f(x) \quad (x \neq v),$$

where the sums run over the lower and upper covers of v in $\widehat{P}$ [10, 12]. The piecewise-linear toggles are its tropicalisation [10]. As in the combinatorial case, each T_v is an involution, and T_u and T_v commute unless u and v are adjacent in $\mathcal{H}(P)$ [12, §2]. These two facts are all we use about toggles in Sections 3–6, so those sections apply verbatim to combinatorial and piecewise-linear toggles.

For a total order $L = (x_1, \ldots, x_N)$ of P put

$$\mathrm{Tog}_L = T_{x_N} \circ \cdots \circ T_{x_2} \circ T_{x_1},$$

so that x_1 is toggled first. *Birational rowmotion* is $\mathrm{Row} = \mathrm{Tog}_L$ for L the reverse of a linear extension, so that higher elements are toggled first. We put

$$K_c = \mathrm{Tog}_{(\beta_1, \ldots, \beta_N)}.$$

Thus Krew_c, $\mathrm{Tog}_{\beta_1 \cdots \beta_N}$ and K_c denote the same word in the toggles, read in the combinatorial, a general, and the birational toggle group, respectively.

3. Toggle orders, orientations and circulation

A total order L of P induces an acyclic orientation O_L of $\mathcal{H}(P)$: each edge points from the element toggled first to the element toggled later.

Lemma 3.1. (a) Tog_L *depends only on* O_L. *We write* $\mathrm{Tog}_O := \mathrm{Tog}_L$ *for any total order* L *with* $O_L = O$.

(b) *Let* x *be a source of an acyclic orientation* O, *and let* O' *be obtained from* O *by reversing all edges at* x, *so that* x *becomes a sink. Then* $\mathrm{Tog}_{O'} = T_x \circ \mathrm{Tog}_O \circ T_x$.

Proof. (a) The total orders inducing O_L are the linear extensions of the partial order generated by O_L. Any two linear extensions of a poset are connected by a sequence of swaps of adjacent, incomparable elements. Two elements that are incomparable in this partial order are not joined by an edge of $\mathcal{H}(P)$, so their toggles commute, and each swap leaves the product unchanged.

(b) Since x is a source, by (a) we may choose a total order $L = (x_1, \dots, x_N)$ with $O_L = O$ and $x = x_1$. As T_x is an involution,

$$T_x \circ \mathrm{Tog}_L \circ T_x = T_x \circ T_{x_N} \circ \cdots \circ T_{x_2},$$

which is Tog of the order $(x_2, \dots, x_N, x)$; this order induces O'. □

For a cycle C of $\mathcal{H}(P)$ with a chosen direction of traversal, the *circulation* $\nu_C(O)$ of an orientation O is the number of edges of C oriented along the direction of traversal minus the number oriented against it. A source-to-sink move does not change any circulation: a cycle through x uses two edges at x, one entering and one leaving x in the direction of traversal, and reversing both swaps their contributions.

Theorem 3.2 (Pretzel [22]; see also [7, Prop. 4.4])**.** *Two acyclic orientations of a connected graph are related by a sequence of source-to-sink moves if and only if they have the same circulation around every cycle.*

Example 3.3. On $[2] \times [2]$, the order $(2,1), (2,2), (1,2), (1,1)$ orients the unique 4-cycle so that, traversed as $(1,1) \to (1,2) \to (2,2) \to (2,1) \to (1,1)$, three edges point against the direction of traversal and one along it. Its circulation is -2, whereas rowmotion's is 0. So the two toggle products are not related by source-to-sink moves. Indeed, the first has infinite order and the second has order 4 [12, Ex. 2.15].

Call $g : P \to \mathbb{Z}$ a *potential* for an orientation O if $g(y) = g(x) + 1$ for every edge $x \to y$ of O. Summing $g(\text{head}) - g(\text{tail})$ around a cycle shows that an orientation with a potential has circulation 0 around every cycle. Conversely, on a connected graph, zero circulation around every cycle allows one to define a potential by integrating along paths. The following corollary needs only the easy direction of Theorem 3.2, and we include a short self-contained proof that also exhibits the conjugating element.

Corollary 3.4. *Let P be a graded poset with connected Hasse diagram. If O_L has circulation 0 around every cycle (equivalently, admits a potential), then* Tog_L *is conjugate to* Row *in the toggle group. More precisely, there are $x_1, \dots, x_k \in P$ with* $\mathrm{Tog}_L = T_{x_k} \cdots T_{x_1} \,\mathrm{Row}\, T_{x_1} \cdots T_{x_k}$.

Proof. Let g be a potential for O_L. In the orientation induced by rowmotion every edge points from the higher to the lower element, so $\varphi = -\mathrm{rank}$ is a potential for it. Along every edge both g and φ change by ± 1, so $g - \varphi$ has constant parity on the connected graph $\mathcal{H}(P)$. Adding a constant to g, which does not change O_L, we may assume that $g \geq \varphi$ and $g \equiv \varphi \pmod 2$.

We modify φ step by step. At each stage φ is a potential for an acyclic orientation O_φ (with $x \to y$ when $\varphi(y) = \varphi(x) + 1$), and Tog_{O_φ} is conjugate to Row by a product of toggles. Suppose $\varphi \neq g$. Let x minimise φ on the set $U = \{v : \varphi(v) < g(v)\}$, so that $g(x) \geq \varphi(x) + 2$. If a neighbour y of x had $\varphi(y) = \varphi(x) - 1$, then $y \notin U$ by minimality, so $g(y) = \varphi(y) = \varphi(x) - 1 \leq g(x) - 3$, contradicting $|g(x) - g(y)| = 1$. Hence every neighbour y has $\varphi(y) = \varphi(x) + 1$, so x is a source of O_φ. Replacing $\varphi(x)$ by $\varphi(x) + 2 \leq g(x)$ turns x into a sink, and by Lemma 3.1(b) it conjugates Tog_{O_φ} by T_x. Each step decreases $\sum_v (g - \varphi)(v)$ by 2, so after finitely many steps $\varphi = g$, $O_\varphi = O_L$, and the claim follows. □

Since circulation is additive, it suffices to verify zero circulation on a set of cycles that spans the cycle space of $\mathcal{H}(P)$ over $\mathbb{Q}$.

Example 3.5. Figure 1 shows the orientation of Example 2.1. The numbers in brackets form a potential, so K_c is conjugate to rowmotion in type A_3 for this c.

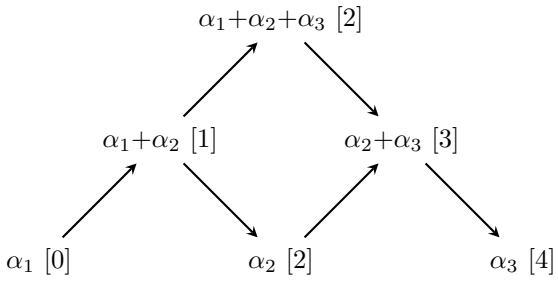


FIGURE 1. The root poset of A_3 with the orientation induced by $w_0(c)$ for $c = s_1s_2s_3$ (each edge points away from the root toggled first) and a potential in brackets.

4. Orientation of cover edges

In this section W is crystallographic. We determine the orientation of each edge $\gamma \lessdot \gamma + \alpha_s$ of $\mathcal{H}(\Phi^+)$ in two steps.

4.1. **Reduction to a comparison with a simple root.** The order $\prec_c$ is a *reflection order* in the sense of Dyer, since it is the inversion order of a reduced word for w_0.

Lemma 4.1 (Dyer [8, (2.2), (2.13)])**.** *If $\alpha \prec_c \gamma$ and $\beta = x\alpha + y\gamma \in \Phi^+$ with $x, y > 0$, then $\alpha \prec_c \beta \prec_c \gamma$.*

Corollary 4.2. *For a cover $\gamma \lessdot \gamma + \alpha_s$, the edge is oriented upward (that is, $\gamma \prec_c \gamma + \alpha_s$) if and only if $\gamma \prec_c \alpha_s$.*

Proof. The root $\gamma + \alpha_s$ is a positive combination of γ and α_s, so by Lemma 4.1 it lies between them in $\prec_c$. If $\gamma \prec_c \alpha_s$, then $\gamma \prec_c \gamma + \alpha_s$; if $\alpha_s \prec_c \gamma$, then $\gamma + \alpha_s \prec_c \gamma$. □

4.2. **The form Ω_c.** Let $(\cdot, \cdot)$ be a W-invariant inner product, let a_{ij} be the Cartan matrix, and put $f_i = (\alpha_i, \alpha_i)/2$, so that $(\alpha_i, \alpha_j) = f_i a_{ij}$. Define a skew-symmetric bilinear form Ω_c on the root lattice by

$$\Omega_c(\alpha_i, \alpha_j) = \begin{cases} -(\alpha_i, \alpha_j) & \text{if } s_i \text{ precedes } s_j \text{ in } c, \\ (\alpha_i, \alpha_j) & \text{if } s_j \text{ precedes } s_i \text{ in } c, \\ 0 & \text{if } i = j \text{ or } a_{ij} = 0. \end{cases}$$

It suffices to treat irreducible W. Then Ω_c is a positive scalar multiple of the form ω_c of Reading and Speyer [24], defined by $\omega_c(\alpha_i, \alpha_j) = -K(\alpha_i, \alpha_j)$ when s_i precedes s_j, where $K(\alpha_i, \alpha_j) = \delta(i)a_{ij}$. Here $\delta(i) > 0$ are the symmetrising constants of [24, §2.2], so that $\alpha_i^\vee = \delta(i)^{-1}\alpha_i$ and K is symmetric. Since both $(\delta(i)a_{ij})$ and $(f_i a_{ij})$ are symmetric, we have $\delta(i)/\delta(j) = f_i/f_j$ whenever $a_{ij} \neq 0$, so $\delta(i)/f_i$ is constant on an irreducible diagram. Note also that Ω_c depends only on c, not on the chosen reduced word.

Lemma 4.3 (Reading–Speyer [24, Prop. 3.11]; Reading [23, Cor. 4.4])**.** *If $\beta \prec_c \beta'$, then $\Omega_c(\beta, \beta') \geq 0$, with strict inequality unless the reflections s_β and $s_{\beta'}$ commute.*

Here Reading's theorem [23, Cor. 4.4] that w_0 is c-sortable is what makes Reading–Speyer's Proposition 3.11 applicable to the inversion order of $w_0(c)$.

Lemma 4.4. *Let $\gamma \lessdot \gamma + \alpha_s$ be a cover with $\Omega_c(\gamma, \alpha_s) \neq 0$. Then the edge is oriented upward if and only if $\Omega_c(\gamma, \alpha_s) > 0$.*

Proof. If the edge is oriented upward, then $\gamma \prec_c \alpha_s$ by Corollary 4.2, so $\Omega_c(\gamma, \alpha_s) \geq 0$ by Lemma 4.3. If it is oriented downward, then $\alpha_s \prec_c \gamma$, so $\Omega_c(\alpha_s, \gamma) \geq 0$, that is, $\Omega_c(\gamma, \alpha_s) \leq 0$. Since $\Omega_c(\gamma, \alpha_s) \neq 0$, its sign determines the orientation. □

Write $\gamma = \sum_t \gamma_t \alpha_t$ and $t \sim s$ for adjacency in the Dynkin diagram. Then

$$\Omega_c(\gamma, \alpha_s) = \sum_{t \sim s} \sigma_{ts} \, (\alpha_t, \alpha_s) \, \gamma_t, \qquad \sigma_{ts} = \begin{cases} -1 & \text{if } s_t \text{ precedes } s_s \text{ in } c, \\ +1 & \text{if } s_s \text{ precedes } s_t \text{ in } c. \end{cases} \tag{1}$$

In Example 2.1, with $(\alpha_i, \alpha_i) = 2$, we get $\Omega_c(\alpha_1, \alpha_2) = \Omega_c(\alpha_2, \alpha_3) = 1$. So the edge $\alpha_1 \lessdot \alpha_1 + \alpha_2$ points up and the edge $\alpha_2 \lessdot \alpha_1 + \alpha_2$ points down, as in Figure 1.

Whenever Lemma 4.4 applies to every cover, the orientation, and hence K_c, depends only on c and not on the reduced word chosen for it. In types F_4, G_2 and H_3 we checked all reduced words directly.

Lemma 4.5. *$\Omega_c(\gamma, \alpha_s) \neq 0$ for every cover $\gamma \lessdot \gamma + \alpha_s$ in types A_n, B_n, D_n and E_n.*

Proof. Simply laced types (normalise $(\alpha, \alpha) = 2$ for all roots): since γ and $\gamma + \alpha_s$ are both roots of norm 2, we get $(\gamma, \alpha_s) = -1$. Expanding, $2\gamma_s - \sum_{t \sim s} \gamma_t = -1$, so $\sum_{t \sim s} \gamma_t$ is odd. By (1), $\Omega_c(\gamma, \alpha_s) = \sum_{t \sim s} \pm\gamma_t$ has the same parity, so it is nonzero.

Type B_n: take simple roots $\alpha_k = e_k - e_{k+1}$ $(k < n)$ and $\alpha_n = e_n$, with $(e_i, e_j) = \delta_{ij}$, so that $(\alpha_k, \alpha_{k\pm1}) = -1$. The covers are

$$e_i - e_j \lessdot e_i - e_{j+1}, \quad e_i - e_n \lessdot e_i, \quad e_i - e_j \lessdot e_{i-1} - e_j, \quad e_i \lessdot e_{i-1}, \quad e_i \lessdot e_i + e_n,$$
$$e_i + e_j \lessdot e_{i-1} + e_j, \qquad e_i + e_j \lessdot e_i + e_{j-1} \ (j - 1 > i).$$

In all but the last family, exactly one neighbour t of s has $\gamma_t \neq 0$, and $\gamma_t = 1$, so $|\Omega_c(\gamma, \alpha_s)| = 1$. In the last family, $s = j - 1$ has the neighbours $j - 2$ and j, with coefficients 1 and 2 in γ, so $|\Omega_c(\gamma, \alpha_s)| \in \{1, 3\}$. □

5. Squares and the cycle space

A *square* of Φ^+ is a quadruple $(\gamma,\ \gamma + \alpha_s,\ \gamma + \alpha_t,\ \gamma + \alpha_s + \alpha_t)$ of positive roots with $s \neq t$. It is a 4-cycle in $\mathcal{H}(\Phi^+)$, consisting of two *s-edges* and two *t-edges*. We write it as $(\gamma;\ s, t)$.

Lemma 5.1. *Suppose that s and t are not adjacent in the Dynkin diagram, and that Ω_c is nonzero on all four edges of the square $(\gamma;\ s, t)$. Then opposite edges of the square are parallel, and its circulation is 0.*

Proof. Since $\Omega_c(\alpha_t, \alpha_s) = 0$, we have $\Omega_c(\gamma + \alpha_t, \alpha_s) = \Omega_c(\gamma, \alpha_s)$. By Lemma 4.4 the two s-edges are therefore both oriented upward or both downward, and likewise the two t-edges. Going around the square, the two s-edges are traversed in opposite directions, and so are the two t-edges, so the circulation is 0. □

In simply laced types, no square has adjacent s and t. Indeed, suppose $s \sim t$, so that $(\alpha_s, \alpha_t) = -1$. Since $\gamma \lessdot \gamma + \alpha_t$, the proof of Lemma 4.5 gives $(\gamma, \alpha_t) = -1$. Hence $(\gamma + \alpha_s, \alpha_t) = -2$, and

$$\|\gamma + \alpha_s + \alpha_t\|^2 = \|\gamma + \alpha_s\|^2 + 2(\gamma + \alpha_s, \alpha_t) + \|\alpha_t\|^2 = 2 - 4 + 2 = 0,$$

which is impossible for a root. In type B_n, the list of covers in the proof of Lemma 4.5 shows that every square has one of the forms

$$(e_i - e_j;\ i-1,\ j), \quad (e_i - e_n;\ i-1,\ n), \quad (e_i;\ i-1,\ n), \quad (e_i + e_j;\ i-1,\ j-1) \ \ (j - 1 > i),$$

and in each, s and t are non-adjacent.

Lemma 5.2. *In types A_n and B_n, the squares span the cycle space of $\mathcal{H}(\Phi^+)$.*

Proof. We embed Φ^+ in the lattice $\mathbb{Z}^2$.

Type A_n. Send $\alpha_i+\cdots+\alpha_j$ to (i,j). The image is the staircase $\Sigma_A=\{(i,j):1\le i\le j\le n\}$. The covers, which add α_{i-1} or α_{j+1}, become exactly the unit steps $(i,j)\to(i-1,j)$ and $(i,j)\to(i,j+1)$ inside Σ_A.

Type B_n. Send $e_i-e_j\mapsto(i,j)$, $e_i\mapsto(i,n+1)$ and $e_i+e_j\mapsto(i,2n+2-j)$. The image is $\Sigma_B=\{(i,p):1\le i\le n,\ i+1\le p\le 2n+1-i\}$, and each family of covers in the proof of Lemma 4.5 becomes a unit step inside Σ_B. Conversely, every unit step between two points of Σ_B arises this way. A step $(i,p)\to(i-1,p)$ adds α_{i-1}. A step $(i,p)\to(i,p+1)$ adds a simple root that depends only on p: it is α_p for $p<n$, α_n for $p\in\{n,n+1\}$, and α_{2n+1-p} for $p\ge n+2$.

In both cases $\mathcal{H}(\Phi^+)$ is thereby identified with the *induced* grid graph on Σ, whose edges are all unit steps between points of Σ. A unit square of the lattice has all four corners in Σ if and only if it is the image of a square of Φ^+. Each region Σ is connected and both row-convex and column-convex.

Consider the plane drawing of this graph. We claim that every bounded face is a unit square. Suppose that a bounded face F contains two adjacent open unit cells. The unit segment between them is not an edge of the graph, so at least one of its endpoints q is not in Σ. This lattice point lies on no edge, so it belongs to F. Since F is bounded, each horizontal ray from q leaves F, so it meets the boundary of F. That boundary consists of edges between points of Σ, and it meets the horizontal line through q only at points of Σ. So Σ contains points on both sides of q in its row, and row-convexity forces $q\in\Sigma$, a contradiction. Hence every bounded face is a single unit cell, whose boundary is the image of a square of Φ^+. For a connected plane graph, the boundaries of the bounded faces span the cycle space, which proves the lemma. □

Proposition 5.3. *In types A_n and B_n, K_c is conjugate to* Row *for every Coxeter element c.*

Proof. By Lemma 4.5, Ω_c is nonzero on every cover. By the discussion after Lemma 5.1, every square has non-adjacent indices, so by Lemma 5.1 every square has circulation 0. By Lemma 5.2 the squares span the cycle space, so every cycle has circulation 0, and Corollary 3.4 applies. □

6. All crystallographic types

Theorem 6.1. *For every finite crystallographic Coxeter group W and every Coxeter element c, the map $\mathrm{Tog}_{\beta_1\cdots\beta_N}$ is conjugate to rowmotion in the combinatorial, piecewise-linear and birational toggle groups of $\Phi^+(W)$. In particular Conjecture 1.1 holds.*

Proof. It suffices to treat irreducible W. As in Proposition 5.3, it suffices to show that the squares span the cycle space of $\mathcal{H}(\Phi^+)$ and that every square has circulation 0. Types A_n and B_n are Proposition 5.3. Homology is taken with $\mathbb{Q}$ coefficients.

Type D_n ($n\ge 4$). Take simple roots $\alpha_k=e_k-e_{k+1}$ $(k<n)$ and $\alpha_n=e_{n-1}+e_n$. The map $e_i-e_j\mapsto(i,j)$, $e_i+e_j\mapsto(i,2n-j)$ sends Φ^+ onto the region

$$\Sigma=\{(i,p):1\le i\le n-1,\ i+1\le p\le 2n-1-i\}.$$

It is injective except that e_i-e_n and e_i+e_n both go to (i,n). Accordingly, $\mathcal{H}(\Phi^+)$ is the grid graph of Σ with the column $p=n$ doubled into two columns n^- and n^+, each joined to the columns $n-1$ and $n+1$.

Let $X^\pm$ be the square complex on Σ that uses the column $n^\pm$. Each is the induced grid complex of a row- and column-convex region, so by the argument of Lemma 5.2 its unit squares span its cycle space, i.e. $H_1(X^\pm)=0$. The unit squares of $X^\pm$ are squares of Φ^+, and the 1-skeleton of $X^-\cup X^+$ is $\mathcal{H}(\Phi^+)$. The intersection $X^-\cap X^+$ consists of two connected, contractible pieces, $p\le n-1$ and $p\ge n+1$. By the Mayer–Vietoris sequence, $H_1(X^-\cup X^+)$ is one-dimensional,

generated by a cycle through $(i, n-1), (i, n^-), (i, n+1), (i, n^+)$. This cycle is the boundary of the square $(e_i - e_{n-1};\ n-1,\ n)$. Hence the squares span the cycle space. Since D_n is simply laced, Lemmas 4.5 and 5.1 apply to every square.

Type C_n. Take $\alpha_k = e_k - e_{k+1}$ $(k < n)$ and $\alpha_n = 2e_n$, with $(e_i, e_j) = \delta_{ij}$, so that $(\alpha_k, \alpha_{k+1}) = -1$ for $k \le n-2$ and $(\alpha_{n-1}, \alpha_n) = -2$. The covers are

$$e_i - e_j \lessdot e_i - e_{j+1}, \quad e_i - e_n \lessdot e_i + e_n, \quad e_i - e_j \lessdot e_{i-1} - e_j, \quad e_i + e_j \lessdot e_{i-1} + e_j,$$

$$e_i + e_j \lessdot e_i + e_{j-1}\ (j-1 > i), \qquad e_i + e_{i+1} \lessdot 2e_i, \qquad 2e_i \lessdot e_{i-1} + e_i.$$

The map $e_i - e_j \mapsto (i, j)$, $e_i + e_j \mapsto (i, 2n+1-j)$, $2e_i \mapsto (i, 2n+1-i)$ identifies $\mathcal{H}(\Phi^+(C_n))$ with the induced grid graph on the region Σ_B of Lemma 5.2, so the squares span the cycle space by the same argument.

On each cover above, the nonzero contributions $|(\alpha_u, \alpha_s)|\gamma_u$ $(u \sim s)$ in (1) form one of the multisets $\{1\}$, $\{2\}$ or $\{1, 2\}$. Hence $\Omega_c(\gamma, \alpha_s) \in \{\pm 1, \pm 2, \pm 3\}$ is nonzero, and Lemma 4.4 applies.

Squares with non-adjacent s, t are handled by Lemma 5.1. The squares with adjacent s, t are exactly $(e_k + e_{k+1};\ k-1,\ k)$.

- For the two s-edges, $\Omega_c(\gamma, \alpha_s)$ and $\Omega_c(\gamma + \alpha_t, \alpha_s)$ both come from the single neighbour k, with coefficients 1 and 2, so they have the same sign.
- For the two t-edges, $|\Omega_c(\gamma, \alpha_t)| = 2$ and $|\Omega_c(\alpha_s, \alpha_t)| = 1$, so adding α_s to γ does not change the sign.

Hence these squares also have parallel opposite edges and circulation 0.

Types E_6, E_7, E_8, F_4, G_2. These are finitely many cases. For every ordering of S we computed $w_0(c)$ and the induced orientation, and verified in exact integer arithmetic that it admits a potential. There are 720, 5040, 40320, 24 and 2 orderings respectively, and every one admits a potential. This suffices by Corollary 3.4. As an independent check, we computed the rank of the square cycles modulo the prime $2^{31} - 1$. Full rank modulo a prime implies full rank over $\mathbb{Q}$, so this confirms that in these types too the squares span the cycle space. We also checked that every square has parallel opposite edges. □

Remark 6.2. Independently of the proofs above, the existence of a potential was verified by computer for every ordering of S in types A_n $(n \le 9)$ and B_n, C_n, D_n $(n \le 8)$.

Remark 6.3. Combining Theorem 6.1 with the theorem of Armstrong, Stump and Thomas [2, Thm. 1.2], the combinatorial map Krew_c has order h if $w_0 = -1$ and $2h$ otherwise, for every finite Weyl group and every Coxeter element c. This extends [6, Cor. 1.6] beyond type A. Moreover Krew_c has the same orbit structure as rowmotion, hence as the Kreweras complement on noncrossing partitions [2], so an equivariant bijection between the two actions exists in every type.

7. The noncrystallographic types

7.1. **Type $I_2(m)$.** Armstrong's root poset of $I_2(m)$ is $\{x, y\} \oplus (z_1 < \cdots < z_{m-2})$: two incomparable minimal elements below a chain. Its Hasse diagram is a tree, so it has no cycles. By Corollary 3.4, every toggle order of it is conjugate to Row. In particular, K_c is conjugate to Row whatever labelling of the poset by roots of $I_2(m)$ is used. It remains to find the order of Row.

Write f' for the labelling after the toggles performed so far. Rowmotion toggles $z_{m-2}, \ldots, z_1$, and then x and y. When x and y are toggled, they receive the new labels $D/(Uf(x))$ and $D/(Uf(y))$, where $D = f(\hat{0})$ and $U = 1/f'(z_1)$ are the same for both. Hence $r = f(x)/f(y)$ is mapped to $1/r$. Put $\Sigma_{xy} = f(x) + f(y)$ and $\Sigma^*_{xy} = 1/f(x) + 1/f(y)$. Then the new values

are $(D/U)\,\Sigma_{xy}^{*}$ and $(U/D)\,\Sigma_{xy}$, so $k = \Sigma_{xy}\Sigma_{xy}^{*} = (1+r)^2/r$ is invariant, and Σ_{xy} is mapped to $kD/(U\Sigma_{xy})$.

Comparing with the toggle rule shows that $(\Sigma_{xy}, z_1, \dots, z_{m-2})$ evolves exactly as birational rowmotion on a chain of $m-1$ elements whose bottom label is kD. That map has exact order m: a chain of $m-1$ elements is $(m-1)$-graded, so this follows from [12, Prop. 7.3]. Since $f(x)$ and $f(y)$ are recovered from r and Σ_{xy}, and $r \mapsto 1/r$ has period 2, the order of Row on $I_2(m)$ is $\mathrm{lcm}(2, m)$.

7.2. **Type H_3.** We use the simple roots of H_3 with $m(s_1, s_2) = 5$ and $m(s_2, s_3) = 3$, and put $\tau = (1+\sqrt{5})/2$. In [5, §3.2, Fig. 2] the 30 positive roots of D_6, written in an orthonormal basis $v_1, \dots, v_6$, are grouped into 15 pairs, one for each positive root of H_3. From each pair take the vector that is lexicographically greater in the coordinates $v_1, \dots, v_6$, and order these 15 roots by the root order of D_6. The result is Armstrong's H_3 root poset: it has $32 = \mathrm{Cat}(H_3)$ order ideals and rank sizes $3, 2, 2, 2, 2, 1, 1, 1, 1$.

Each element is labelled by the H_3 root it represents. For a D_6 root $a = (a_1, \dots, a_6)$ in simple-root coordinates, the vector $(a_5 + \tau a_3,\ a_2 + \tau a_4,\ a_1 + \tau a_6)$ is 1 or τ times a unique positive root of H_3 in simple-root coordinates, and that root is the label of a. Table 1 lists the result. As Williams notes, for H_3 the labelling is part of the data of the problem. In this poset covers are not in general of the form $\beta \lessdot \beta + \alpha_s$ (for example $(1,0,0) \lessdot (\tau, 1, 0)$), so Sections 4 and 5 do not apply, and we argue by computer.

D_6 root		H_3 root	D_6 root		H_3 root	D_6 root		H_3 root
100000	$\mapsto$	$(0,0,1)$	011000	$\mapsto$	$(\tau,1,0)$	111101	$\mapsto$	$(1,\tau,\tau)$
010000	$\mapsto$	$(0,1,0)$	111000	$\mapsto$	$(\tau,1,1)$	111111	$\mapsto$	(τ,τ,τ)
110000	$\mapsto$	$(0,1,1)$	011100	$\mapsto$	$(1,\tau,0)$	111211	$\mapsto$	$(\tau,\tau+1,\tau)$
001000	$\mapsto$	$(1,0,0)$	011110	$\mapsto$	$(\tau,\tau,0)$	112211	$\mapsto$	$(\tau+1,\tau+1,\tau)$
			111100	$\mapsto$	$(\tau,\tau+1,1)$	122211	$\mapsto$	$(\tau+1,2\tau,\tau)$
			111110	$\mapsto$	$(\tau+1,\tau+1,1)$			

TABLE 1. The H_3 root poset inside $\Phi^+(D_6)$ (D_6 roots in simple-root coordinates) and its labelling by positive roots of H_3 (simple-root coordinates).

Conjugacy. There are six orderings of S, giving four Coxeter elements up to commutation. For each, the orientation induced by $w_0(c)$ has circulation 0 around every cycle of a cycle basis, so K_c is conjugate to Row by Corollary 3.4.

Periodicity. Grinberg and Roby report a Sage computation showing $\mathrm{Row}^{10} = \mathrm{id}$ on this poset [12, §21]. We verified this deterministically. Let $x = (x_1, \dots, x_{15})$ be the generic labelling, with boundary labels 1. We computed $F = \mathrm{Row}^5(x)$ and $G = \mathrm{Row}^{-5}(x)$ exactly, as reduced fractions in $\mathbb{Z}[x_1, \dots, x_{15}]$, using multivariate arithmetic with gcd normalisation; the largest polynomial has 11247 terms. The result is $F = G$ identically, so $\mathrm{Row}^{10} = \mathrm{id}$ as rational maps. General boundary labels reduce to $(1,1)$ by the rescalings $f \mapsto \lambda f$ and $f(v) \mapsto t^{\mathrm{rank}(v)} f(v)$, which commute with all toggles. The same computation gives $\mathrm{Row}^5 \neq \mathrm{id}$, and $\mathrm{Row}^2 \neq \mathrm{id}$ at an explicit rational point, so the order is exactly $10 = h$. An independent randomised identity test modulo four large primes agrees.

8. Orders of rowmotion and the proof of Theorem 1.3

By Theorem 6.1 and Section 7, K_c is conjugate to Row, so the two maps have the same order. It remains to recall the order of birational rowmotion on each root poset.

Type A_n. The root poset $\Phi^+(A_n)$ is the triangle $\Delta(n{+}1)$ of Grinberg and Roby [13, Rem. 18.3], on which Row has order exactly $2(n+1) = 2h$ for $n \geq 2$ [13, Cor. 18.9].

Type B_n. The root poset $\Phi^+(B_n)$ is isomorphic to the triangle NEtri$(2n)$ [13, Rem. 19.3], for which Grinberg and Roby conjectured that Row has order $2n$ [13, Conj. 19.5]. This follows from the work of Johnson and Liu [16, Cor. 5.3] on the trapezoid

$$T_{r,s} = \{(I, J) \in [r + s - 1] \times [s] : I - J < r,\ I + J > s\}$$

(with the product order), on which they show that rowmotion has order exactly $r + s$. The map $(i, p) \mapsto (p - 1, n + 1 - i)$ is an isomorphism from the grid model of $\Phi^+(B_n)$ in Lemma 5.2 onto $T_{n,n}$, since it sends up-steps to unit steps. Johnson and Liu state their result for positive real labellings with boundary labels 1. For graded posets, general boundary labels reduce to this case by renormalisation [10], and since $\mathbb{R}^N_{>0}$ is Zariski dense, $\mathrm{Row}^{2n} = \mathrm{id}$ holds as an identity of rational maps. So the order on $\Phi^+(B_n)$ is exactly $2n = h$. The type C_n labelling gives an isomorphic poset.

Type $I_2(m)$. The order is $\mathrm{lcm}(2, m)$ by Section 7. This also follows from [12, Prop. 7.3], as noted in [12, §21].

Type H_3. The order is $10 = h$ by Section 7.

In each coincidental type the order divides $2h$. This proves $K_c^{2h} = \mathrm{id}$ and Theorem 1.3. As a sanity check, floating-point iteration of birational rowmotion at a random positive labelling recovers these orders for A_n $(n \le 5)$, B_n $(n \le 6)$ and $I_2(m)$ $(3 \le m \le 14)$.

Remark 8.1 (Hopkins' conjecture)**.** Hopkins conjectured that birational rowmotion on every root poset of coincidental type has order dividing $2h$ [15, Conj. 4.40]. He noted that Grinberg and Roby had established it for A_n, H_3 and $I_2(m)$ (for H_3 by a computer calculation [12, §21]), leaving type B_n open. As recently as 2023 Grinberg and Roby still described periodicity on all coincidental root posets as open [14]. The results recalled above settle it. The B_n case is due to Johnson and Liu [16], via the isomorphism $\Phi^+(B_n) \cong T_{n,n}$. Our contribution is a deterministic exact certificate in type H_3.

9. Concluding remarks

Remark 9.1 (Related work)**.** The conjugacy of products of involutions whose orders differ by source-to-sink moves is classical for Coxeter elements [25, 11, 17], and for toggle groups it appears in [9, §3], where toggle actions on noncrossing partitions and their homomesies are also studied. For minuscule posets, Okada [19] proved that birational Coxeter-motion is periodic and conjugate to rowmotion. Thomas and Williams [27] developed rowmotion as a sequence of local moves ("slow motion") on trim lattices. What is new here is a uniform description of the toric-equivalence class, in the sense of [7], of the $w_0(c)$ orientation, through the sign of Ω_c.

Remark 9.2 (Open questions)**.** (1) Our proof in types E, F, G and H_3 is computational. A uniform argument would be desirable, for instance one that extends the square analysis of Section 5 to the exceptional root posets and to Armstrong's noncrystallographic posets. (2) In type H_3, K_c depends on the chosen labelling of the root poset by roots. It would be interesting to know which labellings give a map conjugate to rowmotion.

Code

All computations are in the ancillary files of this arXiv submission (Python and C++). The file `anc/README.md` lists, for each computational claim in the paper, the file, the command and the expected output. The files cover:

- the root posets, c-sorting words and orientations (`roots.py`, `orient.py`);
- the exact potential and square checks (`fastcheck.cpp`, `squares.cpp`, `omega.cpp`);
- the H_3 conjugacy check (`h3run.py`);

- the H_3 certificate (`h3_sym_half.py`, with its run log);
- the floating-point sanity checks (`roworder.py`, `i2m.py`).

Use of AI

I used Claude Opus 5.5 (Anthropic) throughout this project: to search the literature, to develop and test the arguments, to write and run the code, and to draft the text. I then had every part checked by separate Claude sessions acting as referees. They re-derived each proof, compared each cited statement with its source, and recomputed the computational claims, in part with code written from scratch. Their reports led to corrections that are included here. These checks were made with the same AI system and do not replace expert refereeing. Every computational claim can be reproduced from the ancillary files. The one statement that rests on a long computation, $\mathrm{Row}^{10} = \mathrm{id}$ in type H_3, is certified by an exact symbolic calculation, not a numerical one. I have read the final manuscript and take full responsibility for its contents; any remaining errors are mine.

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